NAGIREDDY VENKATA SAI KRISHNA

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.19
Sree Kasyap junior college	Class XII (Andhra Pradesh Board of Intermediate Education)	2019	8.96%
Sri Prakash Vidyaniketan	Class X (Central Board of Secondary Education)	2017	10.00%

INTERNSHIPS

RINL - Vizag Steel Plant | Manufacturing & Quality Intern

- Studied **steel casting, heat-treatment and manufacturing processes** with focus on manganese-steel components.
- Performed **casting defect analysis, quality inspection and root-cause identification** for process-related defects.
- Supported **corrective-action analysis and quality-control practices** to improve casting quality and process reliability.

APSSDC - Siemens Center | CNC & CAM Engineering Intern

- Executed **multi-axis CNC machining operations** and developed **G-code/M-code programs** for manufacturing components.
- Generated and optimized **CAM toolpaths** considering feed rate, spindle speed and depth of cut.
- Applied **CNC process planning, machining parameter optimization and manufacturing inspection** for improved process efficiency.

PROJECTS

FEM Project -- Ultrasound-Induced Shear-Wave Generation

- Developed a **2D axisymmetric ABAQUS/Explicit finite-element model** using CAX4R/ACAX4R elements.
- Simulated **focused ultrasound excitation, radiation-force-induced deformation and shear-wave generation** in hyperelastic soft tissue.
- Performed **mesh-resolution analysis** and evaluated deformation/wave propagation characteristics for model validation.

CFD Project -- Mixed Convection over a Heated Backward-Facing Step

- Conducted **12 steady-state RANS simulations** for Re = 5000-11000 and Ri = 0.01, 1, 100 using **ANSYS Fluent**.
- Implemented **k- SST turbulence model** and **Boussinesq approximation** to analyze mixed-convection flow behaviour.
- Evaluated **velocity profiles, recirculation zones and reattachment length** to assess Reynolds/Richardson number effects.

B.Tech Thesis -- Corrosion Behaviour of FSW Aluminium 5083 Joints

- Investigated **friction stir welded (FSW) Aluminium 5083 joints** with different groove geometries and processing conditions.
- Performed **electrochemical corrosion testing** and analyzed corrosion behaviour across **weld nugget, TMAZ and HAZ**.
- Evaluated **corrosion rate and microstructural characteristics** to identify process-property relationships and joint performance.

M.Tech Thesis -- Spatial Light Modulator assisted laser fabrication of copper micro scale structures

- Developed **SLM-based laser beam shaping, steering and multi-spot generation** using Fourier optics and phase-mask design.
- Performed **CCD-based beam characterization**, centroid detection, ROI intensity extraction and **diffraction-efficiency analysis** using MATLAB.
- Investigated **laser-assisted copper material transfer** for precision **micro-circuit/interconnect fabrication** and process optimization.

SKILLS

Production & Industrial Engineering: Industrial Engineering, Operations Research, Production Planning & Control, Manufacturing Systems, Operations Analysis, Optimization Techniques, Engineering Statistics.

Simulation & CAD: SolidWorks, CATIA V5, ANSYS Fluent, ANSYS Mechanical, ANSYS Workbench, ABAQUS.

Manufacturing & Process Engineering: Advanced Manufacturing Processes, CNC Machining, CAM, G-code/M-code, Manufacturing Automation, Process Planning, Process Optimization, Quality Control, Root Cause Analysis, GD&T

RELEVANT COURSES

Mechanical & Materials: Thermodynamics, Strength of Materials, Machine Design, Materials Science, Fluid Mechanics.

Manufacturing & Production: Production technology**,** Advanced Manufacturing Processes, Manufacturing Automation, Advanced Material Joining Processes.

Simulation & Computational: Finite Element Method, Computational Fluid Dynamics, Machine Learning & Its Applications.

POSITIONS OF RESPONSIBILITY

Indian Institute of Production Engineers (IIPE) - Technical Activities

- Conducted technical events and knowledge-sharing sessions for junior students at JNTUK, promoting technical learning and peer engagement.
- Coordinated student activities under IIPE, developing technical event management, communication and team coordination skills.

EXTRACURRICULAR

- Secured the championship title in the Inter-Department Basketball Tournament during B.Tech, demonstrating teamwork, leadership and competitive spirit.
- Represented IIT Hyderabad in Inter-Hostel Basketball events, demonstrating collaboration, discipline and sportsmanship.
- Coordinated the JNTUK University Platinum Jubilee celebrations, gaining experience in alumni engagement, event coordination and professional networking.
- Coordinated a Lathe Machine technical event during the Department Fest, managing participant coordination, event execution and technical activities.

INTERESTS AND HOBBIES

- Sports statistics analysis and strategic game insights.
- Building professional LinkedIn connections within Additive Manufacturing, Laser Processing and Semiconductor Manufacturing.
- Listening to music.